

# Al Naheyan - file encryption manual

**Command:** filecrypt

**Description:** filecrypt is a command that encrypts and decrypts files using a user-specified key and a Vigenere cipher. The command reads the input file in chunks, applies encryption or decryption to each byte using the key, and writes the result into a new output file.

**Syntax:** ./filecrypt -e|-d -k <key> -i <input file> -o <output file>

-e        Encrypt mode  
-d        Decrypt mode  
-k <key>    Encryption/Decryption key  
-i <file>    Input file  
-o <file>    Output file

Encrypt - ./filecrypt -e -k SECRET -i notes.txt -o notes.bin

Decrypt - ./filecrypt -d -k SECRET -i notes.enc -o recovered.txt

## Example:

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Input.txt: HELLO WORLD

./filecrypt -e -k SECRET -i input.txt -o encrypted.bin

File encrypted successfully -> encrypted.bin

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./filecrypt -d -k SECRET -i encrypted.bin -o output.txt

File decrypted successfully -> output.txt

## System calls:

open() - open files

read() - read files

`write()` - write files

`close()` - close file descriptor

`mlock()` - locks key in memory

`munlock()` - unlocks key from memory

`signal()` - gracefully exit on interruptions